

Programme	B. Sc. Mathematics Honours				
Course Code	MAT5EJ308				
Course Title	BASICS OF CLASSICAL MECHANICS				
Type of Course	Major				
Semester	V				
Academic Level	300 - 399				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	4	-	-	60
Pre-requisites	Higher secondary mathematics and physics				
Course Summary	This course offers quick glimpses in to topics is mathematical statistics that can be of interest to a practitioner of science, with special emphasis on statistical inference.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Understand the basic problems that mechanics seeks to solve	Ap	P	Instructor-created exams
CO2	Place the Newtonian formulation of dynamics from high school on a rigorous footing	U	C	Seminar Presentation/ Group Tutorials
CO3	Understand the limitations of the Newtonian formulation of mechanics	Ap	P	Instructor-created exams
CO4	Learn how to improve upon the Newtonian formulation using concepts like Lagrangian & Hamiltonian	Ap	P	Instructor-created exams
CO5	Appreciate how different formulations to the same problem are equivalent but differ in convenience	U	C	Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Textbook		Classical Mechanics: The Theoretical Minimum, Leonard Susskind, Penguin Books, 2014, ISBN 978-0141976228		
Module	Unit	Content	Hrs (48 +12)	Marks (70)
I	Newtonian Dynamics		12	15
	1	Chapter 1: The Nature of Classical Physics		
	2	Chapter 2: Motion		
	3	Chapter 3: Dynamics: Aristotle's Law of Motion		
	4	Chapter 3: Dynamics: Mass, Acceleration & Force		
	5	Chapter 3: Dynamics: Some Simple Examples of Solving Newton's Equations		
	6	Appendix 1: Central Forces & Planetary Orbits		
II	Transition to Advanced Mechanics – Lagrangian Formulation		12	20
	7	Chapter 4: SoMTOP: Systems of Particles, Space of States, Momentum & Phase Space		
	8	Chapter 4: SoMTOP: Action, Reaction & Conservation of Momentum		
	9	Chapter 5: Energy: Force & Potential Energy		
	10	Chapter 5: Energy: More Than One Dimension		
	11	Chapter 6: PoLA: The Transition to Advanced Mechanics, Derivation of Euler-Lagrange Equation		
	12	Chapter 6: PoLA: More Particles & More Dimensions, What's Good About Least Action?		
	13	Chapter 6: PoLA: Generalised Coordinates & Moments, Cyclic Coordinates		
III	The Transition to Hamiltonian Formulation		12	20
	14	Chapter 7: S&CL: Preliminaries, Examples of Symmetries, More General Symmetries		
	15	Chapter 7: S&CL: The Consequences of Symmetry, Back to Examples		

	16	Chapter8: Hamiltonian Mechanics & Time-Translation Invariance: Time-Translation Symmetry, Energy Conservation, Phase Space & Hamilton's Equations		
	17	Chapter8: HM&TTI: The Harmonic Oscillator Hamiltonian, Derivation of Hamilton's Equations		
	18	Chapter 9: The Phase Space Fluid & the Gibbs-Liouville Theorem: The Phase Space Fluid, A Quick Reminder		
	19	Chapter 9: PSF&GLT: Flow & Divergence, Poisson Brackets		
IV	Poisson's Formulation & Electromagnetism		12	15
	20	Chapter 10: PBAMS: An Axiomatic Formulation of Mechanics, Angular Momentum, Levi-Civita Symbol, Back to Angular Momentum		
	21	Chapter 10: PBAMS: Rotors & Precession, Symmetry & Conservation		
	22	Chapter 11: Electric & Magnetic Forces		
V	Open Ended Module		12	
	1	Content from Quantum Mechanics: The Theoretical Minimum, Leonard Susskind & Art Friedman, Basic Books, 2014		
	2	Content from Special Relativity & Classical Field Theory: The Theoretical Minimum, Leonard Susskind & Art Friedman, Basic Books, 2017		
References: 1. Classical Mechanics, John R. Tyler, University Science Books, 2005. 2. Classical Mechanics, 3rd Edition, Herbert Goldstein, Charles P. Poole Jr., John L. Safko, Pearson Education Limited, 2014. 3. Principles of Mechanics, J. L. Synge & B. A. Griffith,				

Mapping of COs with PSOs and POs :

	PSO1	PSO2	PSO3	PSO4	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO 1	3	1	1	3	3	0	1	0	3	1	2
CO 2	3	3	2	3	3	0	1	0	3	1	2
CO 3	3	0	0	3	3	0	1	0	3	1	2
CO 4	3	2	2	3	3	0	1	0	3	1	2
CO 5	3	1	1	3	3	0	1	0	3	1	2

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal Exam	Assignment	Seminar	Viva	End Semester Examinations
CO 1	✓	✓	✓	✓	✓
CO 2	✓	✓	✓	✓	✓
CO 3	✓	✓	✓	✓	✓
CO 4	✓	✓	✓	✓	✓
CO 5	✓	✓	✓	✓	✓